

Model-independence of the raw-Δz generalization (MLP vs trees)

Experiment 06 · ddG_with_Boltz · same corpus/features/splits as 01, MLP instead of gradient-boosted trees.

Re-running the full generalization holdout suite with a **neural-network MLP** in place of gradient-boosted trees reproduces the result on every split — the MLP **matches or slightly beats** the tree model throughout (random 0.80, unseen-protein 0.79, per-protein mean 0.83). Two independent model families land on the same generalization profile, so the result is a property of the **raw-Δz representation**, not of any one estimator.

1. Motivation

The generalization study established that a raw-Δz feature set transfers across proteins, homology, and de-novo designs — using gradient-boosted trees. If a neural network reaches the same numbers on the same splits, the finding is a property of the representation rather than the model, and we gain a second, interchangeable regressor.

2. Methods

Identical corpus (Tsuboyama, 12,359 mutations / 412 proteins), identical 256 raw-Δz features, and identical holdout definitions to the tree-based study. The only change is the regressor: a five-seed ensemble of MLPs (median-impute → standardize → network), averaged to reduce variance. Pooled out-of-fold Pearson r is reported per holdout, plus a homology sweep at 30/50/90% sequence identity.

3. Results

Holdout	Trees (01)	MLP	Δ
Random (10-fold)	0.783	0.803	+0.020
Protein (unseen proteins)	0.774	0.792	+0.018
De-novo (natural ↔ designed)	0.705	0.703	-0.002
Substitution (leave-one-out)	0.772	0.765	-0.007
Source residue	0.754	0.770	+0.016
Target residue	0.743	0.761	+0.018
Chemistry class	0.734	0.732	-0.002
Homology 30 / 50 / 90 %	0.765 / 0.766 / 0.772	0.781 / 0.785 / 0.790	+~0.017

Table 1. Pooled Pearson r per holdout — trees (experiment 01) vs MLP.

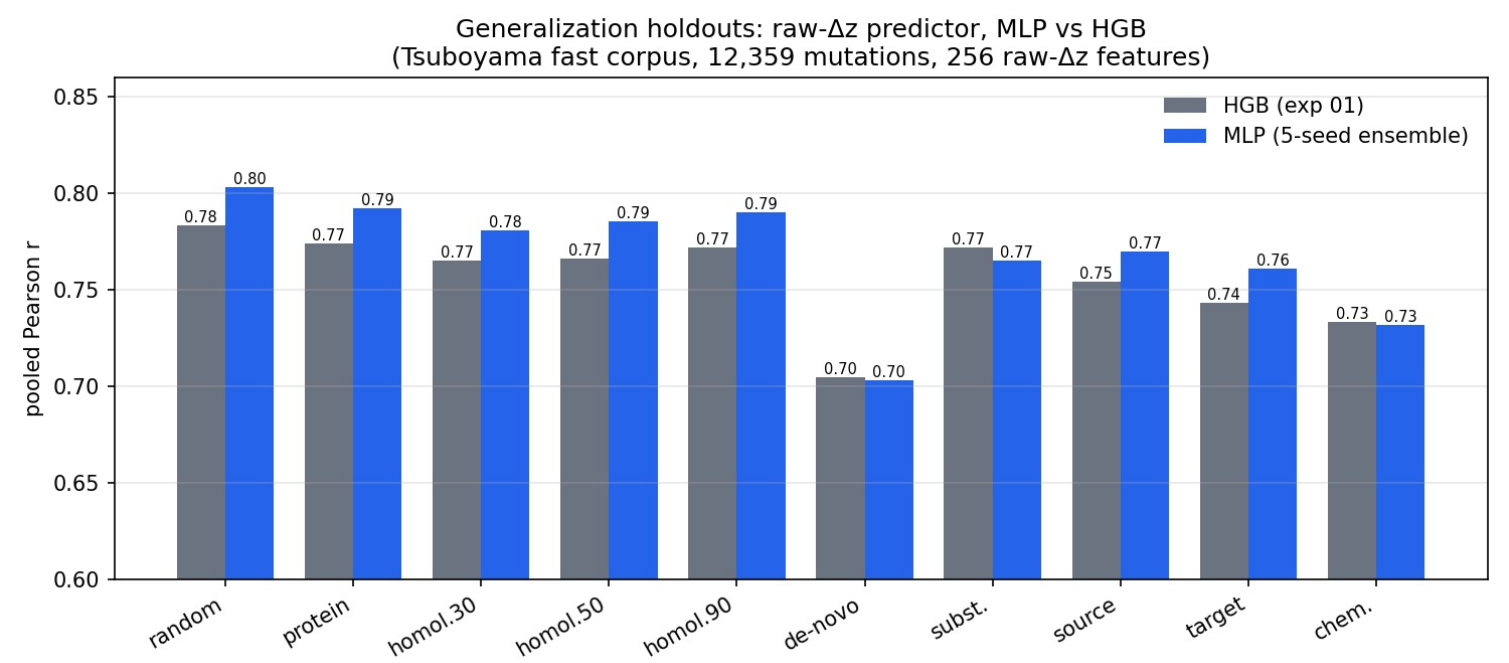


Figure 1. Pooled Pearson r per holdout, MLP vs the tree baseline.

4. Error is set by training density, not the model

Binning the predictions by measured ΔΔG exposes the shared limitation: a regression-to-mean bias (over-predicts low ΔΔG, under-predicts high) with error minimized in the dense centre and rising toward both tails. Relating per-bin error to the **training observation density** in ΔΔG space, error is almost perfectly anti-correlated with density (Spearman ρ = −0.97 over bins). The accuracy at a given ΔΔG is governed by how densely the training set sampled that value — a coverage effect, identical across model families.

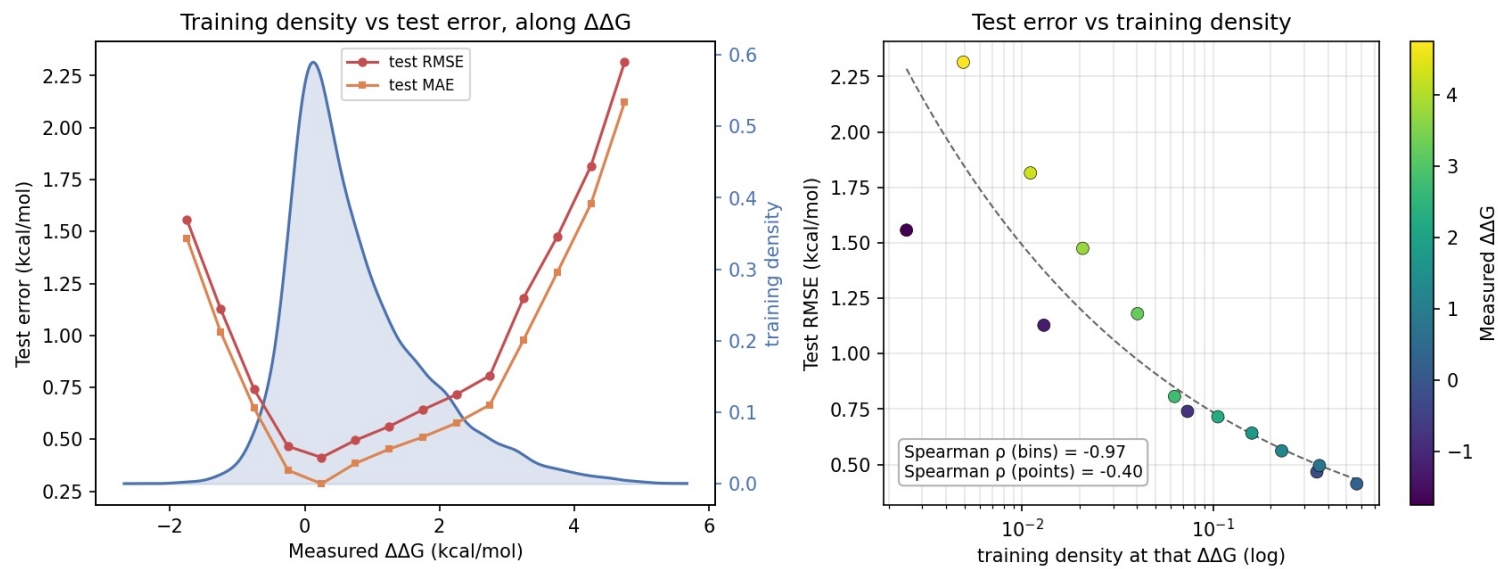


Figure 2. Test error vs training density in $\Delta\Delta G$ space. Left: density and error are mirror images along $\Delta\Delta G$. Right: error falls monotonically with training density.

5. Conclusion

The MLP is statistically even with the tree model on every holdout, including the hardest transfers. The generalization is a property of the raw- Δz features, and the residual weakness (under-prediction of large effects) is a training-coverage effect rather than a modeling choice.